EMPLOYEE ATTRITION ANALYSIS

Short Project Report

1. Introduction

Employee attrition is an important HR problem that can lead to loss of skilled employees, reduced productivity, and increased recruitment costs. This project uses HR analytics and data visualization to identify patterns and factors associated with employee attrition.

2. Objectives

- Analyze employee attrition patterns.
- Identify important factors contributing to attrition.
- Study departmental and demographic trends.
- Examine tenure, job role, salary, satisfaction, and work-life balance.

3. Data and Methodology

The analysis considers attributes such as age, gender, department, job role, monthly income, job satisfaction, work-life balance, years at company, overtime, performance rating, and attrition status. The preprocessing flow includes data collection, cleaning, missing-value handling, transformation, and analysis.

Python tools such as Pandas, NumPy, Matplotlib, and Seaborn are used for data analysis and visualization.

4. Visualization and Findings

Exploratory data analysis focuses on attrition rate, department and job-role patterns, tenure, age, salary, satisfaction, and work-life balance. The project presentation highlights higher attrition risk during the first two years of employment. Low salary, poor work-life balance, low satisfaction, and frequent overtime are also identified as important factors in the analysis.

5. Conclusion

Employee attrition analysis helps organizations understand workforce risk and make evidence-based HR decisions. Visualization makes these patterns easier to understand and supports targeted retention strategies, compensation review, employee engagement programs, and workforce planning.

6. Future Scope

Future work can include AI-based attrition prediction, real-time employee engagement monitoring, predictive risk modelling, HR dashboards, and machine-learning models for prioritizing retention actions.